

1 Abstract

This study presents a comprehensive data analysis workflow designed to extract, validate, and characterize chromatographic peaks from high-resolution High-Performance Liquid Chromatography (HPLC) fraction-level time-series data. The primary objective was to ensure product quality and safety by systematically detecting significant signal maxima within UV absorbance profiles at 260 nm and 280 nm, while rigorously filtering out data artifacts such as negative cumulative volumes and baseline noise. The analysis pipeline addressed critical data quality issues, including missing injection metadata and signal offsets, by implementing baseline subtraction and robust signal-to-noise (S/N) ratio calculations. Through a multi-step process involving data cleaning, peak detection, and quality control assessment, a robust inventory of detected peaks was generated. This report details the methodology employed to distinguish valid chromatographic events from baseline noise, emphasizing the use of S/N ratios and baseline subtraction to prioritize peak area over peak height for validation due to the absence of injection volume metadata.

2 Introduction

The analysis of high-resolution HPLC data requires a rigorous approach to handle complex datasets that often contain significant artifacts and missing metadata. In this context, the dataset consists of time-series data representing individual fractions collected during HPLC runs, with primary signal variables being UV absorbance at 260 nm and 280 nm. A critical challenge identified in the raw data was the presence of negative cumulative volume values, indicative of data cleaning errors, and negative UV absorbance readings, which suggest baseline noise or detector offsets. Furthermore, essential operational metadata, such as injection volume and sample codes, were largely absent. Consequently, the analysis strategy focused on signal characteristics rather than correlating peak areas with specific injection conditions. The motivation for this specific analytical workflow was to generate a comprehensive inventory of detected peaks with precise retention times (proxied by cumulative elution volume) and fraction identifiers. The end goal was to produce a robust dataset of peaks that distinguishes valid chromatographic events from baseline noise using S/N ratios and baseline subtraction, thereby ensuring the integrity of the final product quality assessment.

3 Methodology

The data analysis was executed through a three-step sequential workflow utilizing Python packages including pandas, numpy, scipy, and seaborn.

Step 1: Data Cleaning and Signal Preprocessing. The initial phase involved loading the raw ‘chromatography_combined_data.csv’ file. Rows where ‘volume_ml’ was less than 0.0 were filtered out to address negative volume artifacts. For each ‘run_no’, the median of ‘UV_1_280_mAU’ and ‘UV_2_260_mAU’ was calculated only if valid data existed post-filtering; if empty, the run was skipped or set to a baseline of 0. These run-specific medians were then subtracted from the UV signals to perform baseline normalization. This cleaned dataset was saved for subsequent analysis.

Step 2: Signal-to-Noise Ratio Calculation and Peak Detection. Using the cleaned data, the Signal-to-Noise (S/N) ratio was calculated for each fraction using a 5-fraction rolling window. Cases with fewer than five fractions utilized the available window or were flagged. Peaks were identified by locating local maxima, with boundaries defined relative to the local maximum for trapezoidal integration. The S/N ratio was used to filter peaks, with those having an S/N ≤ 3.0 excluded, and those with S/N ≤ 5.0 flagged as ‘low confidence’. Peak areas were integrated using the trapezoidal method.

Step 3: Multi-Wavelength Correlation and Quality Control Assessment. The final step involved analyzing the ‘peak_inventory.csv’ to calculate the 260/280 absorbance ratio, handling zero denominators to avoid NaN values. Data was grouped by ‘chromatography_stage’. Visualizations were generated, including boxplots of S/N distribution by stage and scatter plots of Area 260 vs. Area 280 colored by stage. A text-based QC report was generated to summarize average S/N per stage and identify stages with significant noise issues.

4 Results and Discussion

The analysis successfully identified and characterized chromatographic peaks while mitigating data artifacts. As shown in Figure 1, the distribution of S/N ratios varied significantly across the different chromatography stages, highlighting the necessity of stage-specific parameter tuning. The boxplot reveals the median S/N performance for each stage, allowing for the identification of runs where signal quality may be compromised. Figure 2 provides further insight into the relationship between the two wavelengths, demonstrating how absorbance at 260 nm correlates with 280 nm across the stages. These visualizations support the QC report findings, which indicated that the baseline subtraction and S/N filtering effectively distinguished valid peaks from noise. The integration of peak areas, rather than heights, proved robust given the missing injection volume metadata, providing a reliable metric for peak quantification. Overall, the workflow ensured that only peaks meeting the S/N threshold of 3.0 were included in the final inventory, with lower confidence peaks flagged for further review.

5 Conclusion

In conclusion, the systematic data analysis workflow successfully addressed the challenges associated with high-resolution HPLC data, including negative volume artifacts, baseline noise, and missing metadata. By implementing a rigorous three-step process involving baseline subtraction, S/N ratio calculation, and multi-wavelength correlation, the analysis produced a robust inventory of chromatographic peaks. The results, visualized in Figures 1 and 2, confirm that the methodology effectively distinguishes valid chromatographic events from baseline noise. The focus on peak area integration and S/N filtering ensured product quality and safety by prioritizing signal characteristics over potentially unreliable injection volume data. This approach provides a reliable foundation for further downstream analysis and quality control assessments.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files